

QuantumNet Solutions SL

Tender: CTTI-2026-36

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QuantumNet Solutions SL presents this technical proposal for CTTI tender CTTI-2026-36. Our offer is grounded in direct operational experience delivering QKD infrastructure across two completed public-sector pilots in Catalonia and the Basque Country.

1.1 PLA DE MIGRACIO — PROJECT TEAM

The project is led by Dr. Jordi Puig, Programme Manager, holding CCDP certification with 7 years optical network experience and 3 years focused exclusively on CV-QKD deployments. He served as lead project manager on both public-sector QKD pilots referenced above, overseeing node deployment, acceptance testing and handover at each site.

Solution architecture is the responsibility of Maria Sola, Azure-certified Network Architect with 6 years complex optical infrastructure design and 2 years embedded at Telefonica Barcelona research node integrating QKD into carrier-grade DWDM. Two CCNP-certified Network Administrators with 3+ years QKD-adjacent experience complete the migration team. A CCIE-certified Level 3 engineer holds formal manufacturer accreditation and commits 120 dedicated hours.

1.2 EXECUCIO CRITICA I DESPLEGAMENT

Team for Execution Phase

The execution team mirrors the migration team qualifications. Security engineering is handled by a dedicated engineer with 4 years applied cryptography, FIPS 140-3 HSM deployment experience, and ISO 27001 Lead Implementer certification.

Risk Identification

Risk 1 — Optical signal degradation at legacy fibre nodes: Several nodes may present aged fibre runs compressing quantum channel margin below operational threshold. Mitigation: structured pre-installation OTDR survey at each node minimum 3 weeks before equipment delivery, with defined acceptance criterion for fibre qualification. Nodes failing qualification undergo corrective fibre assessment before QKD commissioning.

NOTE: Only one risk has been identified and documented with a specific mitigation plan for this submission.

Service Monitoring Tool

A fully operational NOC dashboard is live from day one of service. It presents Secure Key Rate, excess noise, optical loss, and received signal power per node in real time, refreshed on a 5-minute cycle. All KPIs from PPT Section 3.1.5 are instrumented. Native integration with CTTI SNMP and HTTP infrastructure. Automated alerting with configurable threshold-based notifications routed to both QuantumNet NOC and CTTI simultaneously.

Documentation

Technical manuals will be delivered for all supplied equipment. Documentation packages will be provided in Catalan as required. Full handover documentation will be available at close of service.

Equipment — Flexibility

Full DWDM-to-dark-fibre reconfiguration capability via software-defined wavelength adjustment requires no hardware replacement when network topology changes. Emitter and receiver units replace independently; factory-paired units requiring full system replacement are absent from our offer.

Equipment — Technical Specifications

CV-QKD system delivers minimum 1.5 kbps net symmetric SKR at 10 dB optical loss, exceeding the 1 kbps minimum. Security parameter $\epsilon = 10^{-10}$, an order of magnitude tighter than the required 10^{-9} . KMS component is FIPS 140-3 certified, fully compliant with ETSI GS QKD 004 and 014. Equipment ships in standard 19-inch rack format, 2U per main unit, max depth 550mm.

Equipment — Robustness and Resilience

Dual redundant power supplies on all primary units. Automatic failover with 30-second recovery time. Critical optical components are dual-sourced from two independent manufacturers, eliminating single-vendor dependency.

1.3 ANALISI DE LES DADES — DATA ANALYSIS TEAM

A dedicated two-person data analysis team with specific quantum network traffic analysis experience is assigned to the data analysis phase. Both analysts have previously worked on traffic characterisation in QKD-enabled optical networks and are familiar with the interplay between quantum channel performance metrics and classical traffic coexistence. Deliverables will be structured according to PPT Section 5.4 templates.

1.4 PLA DE DEVOLUCIO DEL SERVEI

The service handover is structured as a 2-month programme beginning 2 months before contract end. A dedicated transition team of two engineers and the solution architect is assigned exclusively to transition activities.

Knowledge transfer comprises four structured sessions: architecture overview (full day), security controls and key management (half day), operational procedures and runbook walkthrough (full day), and vendor management and licence renewals (half day). All sessions delivered in Catalan. A 12-month post-handover warranty support period is included with a P1 response SLA of 4 hours. Complete runbooks and architecture documentation committed for delivery 30 days before handover period begins.

ADVISORY: This document is a synthetic technical proposal generated for evaluation agent testing purposes only. All company names, personnel, certifications, and project references are fictional.